

Krishnaa Vadivel | Computational Physics | Distributed Computing | Machine Learning

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Profile

Computational physics PhD researcher with training across condensed matter, numerical methods, distributed scientific software, and machine learning for accelerated calculation. Applies those tools in first-principles studies of self-trapped excitons and excitonic polarons, many-body theory for electron-phonon coupled systems, and scalable implementations on national-lab HPC platforms.

Research Experience

Yale University, Electrical Engineering

Research Assistant / PhD Researcher

2021–present

- Developed distributed scientific software in Python and C++ with MPI, OpenMP, PETSc, and SLEPc for large linear-algebra and simulation workloads as part of an INCITE award using exascale resources on national-lab HPC systems.
- Developed many-body and field-theoretic formulations of coupled electron, exciton, and phonon systems, including coherent-state and path-integral approaches for time-dependent and self-localized states.
- Ported and optimized computational workloads for accelerator-enabled HPC environments using CUDA and ROCm, with experience running on Frontier at OLCF, Perlmutter at LBNL, and other leadership-class clusters.
- Used PyTorch, PyTorch Geometric, and equivariant graph neural networks to accelerate self-trapped-exciton calculations and related materials simulations.
- Packaged scientific applications in Docker containers for deployment across different computing environments and used TotalView, Linaro DDT, Bash, and Linux command-line tools to debug distributed applications.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and presented APS talks in 2024, 2025, and 2026 on self-trapped exciton formation and related excited-state materials physics.

Professional Experience

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Built CUDA-based data-processing pipelines and custom GPU-enabled engineering tooling for large acoustic waveform datasets generated by downhole logging tools.
- Built full-stack server applications with C#, ASP.NET, and Microsoft SQL Server for database design, real-time remote tool testing, and calibration of temperature and pressure sensors.
- Applied finite-element analysis with dolfinx for frequency analysis of steel pipes in well environments to support in-house physics and engineering studies.

FindLight

Photonics Technical Writer

2018–2019

- Produced technical articles and product-facing photonics content covering lasers, optics, and optoelectronic components for a specialized engineering readership.

Selected Projects

Distributed First-Principles Exciton-Phonon Simulation

- Developed distributed first-principles calculations for excited-state and exciton-phonon problems, with an emphasis on self-trapped excitons, many-body theory, and scalable materials simulation.

Equivariant Graph Neural Networks for Molecular Systems

- Developed PyTorch Geometric equivariant models for molecular-property prediction and explored their use for accelerating scientific calculations.

Agentic Scientific Input Generation

- Built retrieval-driven tooling that reads software documentation and research papers to generate structured computational-science inputs and starter configurations for physics and chemistry codes.

Integrated Photonics and FDTD Modeling

- Designed photonic structures including nanophotonic ring-resonator band-pass filters in telecom bands with Meep-based FDTD simulation.

DOLFINx Finite-Element Analysis

- Built 2D DOLFINx/PETSc simulations for transient heat diffusion, Poisson potential around a circular cutout, and incompressible flow past an immersed cylinder, with PyVista and Matplotlib results.

Education

Yale University

PhD, Electrical Engineering

2021–present

Ongoing doctoral research in computational materials, many-body theory, and scientific computing.

Yale University

MPhil, Electrical Engineering

2023

Yale University

MS, Electrical Engineering

2023

Cornell University

MEng, Electrical and Computer Engineering

2017–2018

Included work in stochastic computing, device-oriented EE topics, and advanced design coursework.

Cornell University

BS, Electrical and Computer Engineering

2013–2017

Included embedded systems, robotics, and core mathematics, physics, and computing foundations.

Selected Publications

2025: de Paula, A. M., Li, S., Hou, B., Vadivel, S., Teles-Ferreira, D. C., Iudica, A., Kabacinski, P., Hosseini, H., McArthur, J., Cerullo, G., et al. (2025). Time-domain observation of ultrafast self-trapped exciton formation in lead-free double halide perovskites. *Journal of the American Chemical Society*, 147(32), 28923–28931.

Selected Conference Presentations

2026: Vadivel, S., Grande, R. D., Strubbe, D. A., & Qiu, D. Y. (2026). First-principles study of exciton-polarons and self-trapped excitons. *APS Global Physics Summit 2026*.

2025: Vadivel, S., Qiu, D. Y., Grande, R. D., & Strubbe, D. A. (2025). First-principles study of self-trapped exciton formation in double perovskites. *APS Global Physics Summit 2025*.

2024: Vadivel, S., & Qiu, D. (2024). First-principles study of self-trapped exciton formation in double perovskites using excited state forces. *APS March Meeting Abstracts 2024*, F01–007.

Selected Coursework

Mathematics: Differential Geometry; Abstract Algebra; Honors Mathematical Analysis II

Computer Science: Theoretical Computer Science; Probabilistic Machine Learning

Physics: Graduate Quantum Mechanics I–II; Solid State Physics I–II; Many-Body Quantum Physics

Electrical Engineering: Semiconductor Physics; Thin Film Science; Lasers and Optoelectronics; Integrated Photonics; VLSI Design; Analog VLSI Design; Mathematics of Signals and Systems; Advanced Embedded Systems

Technical Skills

Languages: Python, C, C++, CUDA, JavaScript, Bash, PowerShell, SQL, C#

Scientific Computing: MPI, OpenMP, PETSc, SLEPc, NumPy, pandas, SciPy, SymPy, matplotlib, mpi4py, petsc4py, slepc4py, dolfinx

Machine Learning: PyTorch, PyTorch Geometric, scikit-learn, equivariant graph neural networks

Systems: Linux command line, macOS, Windows, CMake, Docker, Docker Compose, Kubernetes, gcloud, ASP.NET, Microsoft SQL Server

Debugging / Platforms: TotalView, Linaro DDT, distributed-application debugging, Frontier, Frontera, Perlmutter, and related national-lab or university HPC environments